Whiskey Recommender System

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Dataset

Motivation

After exploring some interesting datasets to study from Kaggle, Professor Julian McAuley's datasets list, and Yelp's dataset, we could not find one that really piqued our interest. So, we decided to take on the challenge and created our own dataset through web scraping. We settled on collecting whiskey datasets from *www.whiskybase.com*, the biggest whiskey database website since 2007 devoted to whiskey enthusiasts[1].

Specification

We split the data we extracted into four files for scalability reasons: rating.json, review.json, whiskey.json, and distillery.json. We wanted to minimize information redundancy in our dataset, which is why we decided to separate the product details (master data) from the user feedback (transaction data). We had two kinds of master data: whiskeys and distilleries, hence whiskey.json and distillery.json. As for the transaction data, *www.whiskybase.com* contains two types of user feedback: numerical ratings only, and ratings with review texts. This is why we also decided to separate the two types of feedback into rating.json and review.json.

whiskey.json	
id	unique whiskey ID
name	name/brand of the whiskey
label	label/sub-brand of the whiskey
distillery	name of the distillery, which produced the whiskey cask
bottler	name of the bottler. Distilleries that bottles their own casks are listed as 'Distillery Bottling'
series	name of the bottling series / commemorative series
category	category of the whiskey
vintage	year the whiskey was distilled, i.e. put into the cask
bottled	year the whiskey was bottled, i.e. moved from cask to the bottle
stated_age	number of years the youngest whiskey in the blend is aged in the cask
casknumber	serial/batch number of the cask the whiskey was aged in
casktype	type of the cask the whiskey was aged in, i.e. type of wood, size, and number of prior fills
strength/abv	alcohol content, measured in ABV (alcohol by volume)
size/ml	list of bottle volumes available, measured in ml (milliliters)
price/eur	price of the whiskey in Euro
added_on	timestamp when the whiskey review was added and the name

	of the submitter

review.json	
whiskey_id	ID of the whiskey being reviewed
user	username of the reviewer
rating	rating of the whiskey
message	general review of the whiskey
nose	nose/aroma review of the whiskey
taste	taste/palate review of the whiskey
finish	finish/aftertaste of the whiskey

rating.json	
whiskey_id	ID of the whiskey being reviewed
user	username of the reviewer
rating	rating of the whiskey

distillery.json	
distillery	name of the distillery, which produced the whiskey cask
country	country the distillery is located in

Exploratory Analysis

Unique Values

We counted the number of unique values of the fields in the various files:

rating.json (1,153,405 entries)	
user	18,779
whiskey_id	75,235

review.json (162,013 entries)	
user	5,726
whiskey_id	47,847

From these two tables, and also considering that rating.json contains 1,153,405 entries while review.json contains 162,013 entries, we can see that significantly more users engage in rating whiskeys compared to reviewing whiskeys. Furthermore, the number of whiskeys with ratings (75,235) is more than 50% bigger than the number of whiskeys with reviews (47,847). This is in line with our expectations that writing a review is a much more involved and time-consuming activity compared to giving a rating.

whiskey.json (76,129 entries)	
id	76,129
name	37,530
label	4,295
series	13,267
category	15
casktype	9,674
distillery	946
bottler	907

Additionally, we could see from the whiskey table that a significant amount of whiskeys with different IDs have the same name. After further analysis, this is typically due to whiskeys from different casks and/or bottling dates being considered different whiskeys despite them having the same name, especially with vintage whiskeys.

distillery.json (1,593 entries)

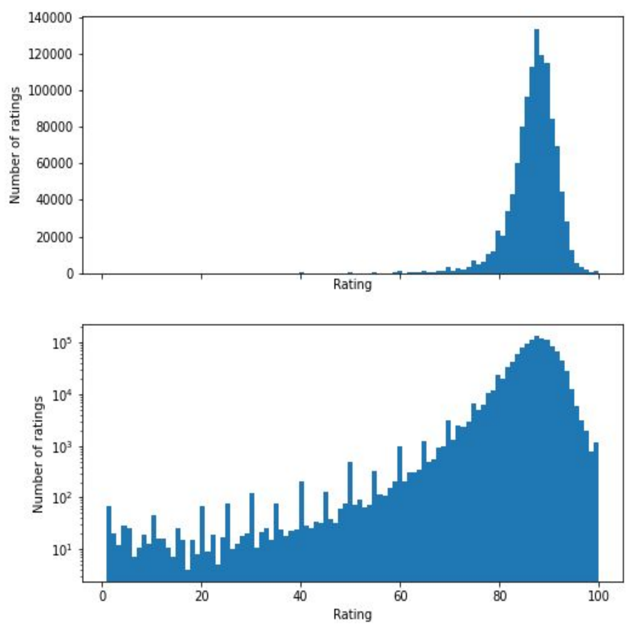
distillery	1,593
country	54

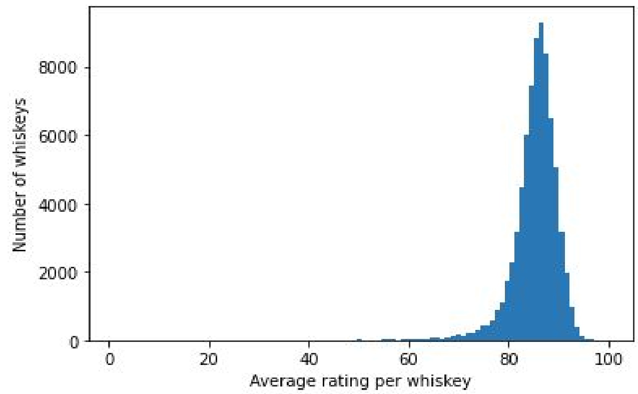
Basic Statistics

Mean	87.148
Mode	88
Median	88
Standard Deviation	5.218
Interquartile Range	5

Distribution of Ratings

We also analyzed the distribution of ratings and average rating per whiskey. More specifically, we analyzed rating distribution globally, as well as average rating per whiskey.





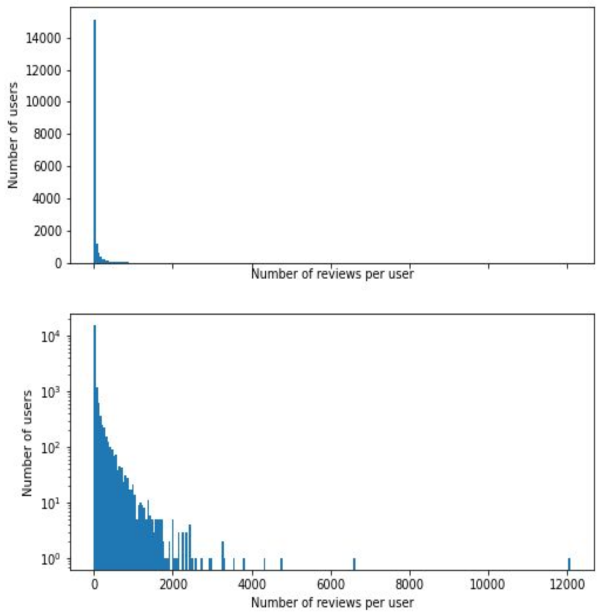
From the figures above, we can see that the majority of the ratings have a value between 70 to 100, and centered around 88. This high center means that raters in general tend to give a high rating, even to whiskeys they don't really like (in comparison to other whiskeys in the dataset).

Furthermore, we can see that the distribution of ratings globally and the distribution of average ratings per whiskey are very similar, roughly matching each other. This means that an unbiased estimator for one distribution can describe the other distribution relatively well, vice versa.

Additionally, one interesting finding is that users tend to give ratings on a multiple of 5. This can be seen on the rating histogram with the logarithmic scale as "protrusions" from the otherwise relatively smooth curve on a regular interval. This is a fact that might be useful to take into consideration when designing models.

Number of Ratings per User

We plotted the number of times each user submitted a rating in total as a histogram.



From this figure, we can see that most of the users have reviewed very few whiskeys; we had to use the logarithmic scale on the y-axis in order to see any appreciable features. In fact, there are 5,057 users who have only submitted one review, 1,876 who have submitted two reviews, and 1,132 who have submitted three reviews. If we exclude users who have submitted less than 10 reviews, we will only have 39.2% of the users remaining but still retain 97.1% of the reviews.

Predictive Task

Motivation

For this assignment, we will be building a rating predictor to determine how likely a user will enjoy new whiskeys. More specifically, our model predicts how much a user will enjoy a new whiskey by predicting the rating that the user will give to the whiskey, especially in relations to other ratings. We can then turn this predictor into a recommender system that recommends whiskey based on which whiskey ranks the highest for a particular user.

Our motivation for choosing this specific task arose from our personal curiosity of building a recommender system using real and up-to-date whiskey data. We want to have more personalized recommendations for whiskey drinkers rather than just relying on the whiskeys' overall ratings.

Data Preprocessing

For this predictive task, we decided to scrape the data we need directly from a whiskey database website to obtain all the relevant features we may need, particularly users' reviews and ratings for every whiskey. This feature will help us identify the correlation between users and whiskeys. During scraping, we performed our first stage of data cleanup: excluding whiskey pages with no users' reviews and ratings specified. This quite significantly speeds up the scraping process and results in a more informative dataset.

As mentioned during our exploratory analysis, since the majority of the users are not active in giving ratings, we decided to further process our data by excluding users who have submitted fewer than 10 reviews. This action dropped 60.8% of the users while still retaining 97.1% of the reviews.

Furthermore, the scraped dataset contained two fields with varying units: strength and price. Whiskey strength was measured by both ABV (alcohol by volume) and proof; since ABV was the dominant unit in the dataset, we decided to convert proof into ABV. Moreover, there were varying currencies used to measure price, including EUR, GBP, USD, AUD, CHF, etc; since the dominant currency was EUR, we decided to convert all currencies to EUR according to the mid-market interbank exchange rate effective on 5 Dec 2020.

Model Evaluation

For our model evaluation pipeline, we will shuffle all 1,153,405 entries in the rating datasets extracted and split 80% : 10% : 10% for training, validation and testing sets respectively. By doing so, we will obtain 922,725 entries for training, and 115,340 entries for both validation and testing sets.

To evaluate the performance of our model, we will be using Root Mean Square Error (RMSE) as our metric. RMSE – the square root of MSE – measures the average magnitude of the errors, i.e. the difference between the predictions and the actual labels. Thus, the lower the RMSE, the better is our model, because it means that our model on average produces smaller errors.

$$RMSE = \sqrt{\tfrac{1}{N}\|y - X\theta\|_2^2} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - X_i \cdot \theta)^2}$$

Models

For this predictive task, we have considered various models. First is the baseline. We have considered 4 naive baseline models, which will be explained in greater detail in the next subsection.

With the goal of creating a recommender system, we initially thought of reframing the problem as a classification problem such that given some combination of whiskey features, if the predicted rating given by the user is above a certain threshold, then we include the whiskey into the recommendation list; otherwise, exclude it from the list. Thus, we initially considered constructing our model using classifiers such as SVM and logistic or softmax regression.

However, converting this problem to a classification problem might be poorly motivated, because a binary outcome can only indicate whether the whiskey is above or below the threshold. Thus, this type of recommender system can only present the user with a list of whiskeys they might enjoy in general, but is unable to rank the whiskeys within the list. A whiskey that the user only slightly likes is in the same list as one that the user enjoys very much, without any distinction between the two.

For this reason, we would instead frame the problem as a regression problem in order to provide a more fine-grained control over the recommendations. Since the recommender system has the numerical rating values to work with, it can sort the whiskey bottles according to the rating and can present just the few couple whiskey bottles users might like the most. For this reason, we will be exploring regression predictive modelings such as linear regression, Ridge regression. These models will be explored in greater detail in the next section.

Baselines

In order to assess the usefulness of our predictor model, we aim to beat the baseline model that we initially set for comparison which has the best RMSE among other baseline alternatives. Some relevant baseline models that we take into account are:

Model	RMSE
Avg. rating globally	4.812
Avg. rating of a particular whiskey	4.034
Avg. rating given by the user	4.808
Avg. rating of all whiskeys from the same country	5.097
Avg. rating of all whiskeys from the same distillery	4.829

These baseline models rely on statistical information of the training sets. The first model simply returns the global rating average for all predictions. For the second model, we predict the rating to be the average rating of a particular whiskey irrespective of the user, and take the global average rating of 87.148 if the whiskey has never been reviewed before (not found in the training sets). On the opposite, the third model predicts the rating to be the average rating given by that specific user regardless of the whiskeys being reviewed, and also takes the global average rating of 87.148 if the user has never reviewed any whiskeys before (not found in the training sets). The fourth and fifth models take the overall rating of all whiskeys based on country and distillery respectively. Out of these 5 models, we picked the one with the lowest RMSE to be our baseline, that is using the average rating of a particular whiskey with RMSE of 4.034.

Model

Our Model

As mentioned in the previous section, we examined various models related to regressing numerical values, i.e. predicting the actual ratings. Therefore, we also performed features selection by iterating through all subsets of the dataset fields as well as exploring other recommender system techniques like Jaccard Similarity and Bag-of-Words, and choosing the one with the lowest RMSE.

Combination of Features

Our first approach is to combine various features in the dataset. We used Ridge regression so as to regularize the parameters between the different features. We also tried linear regression without regularization, but the majority of the models did not generalize well, and there were a few instances where the MSE rose to about 10^{12}.

The first feature we included was the average rating of whiskeys, because from looking at the baseline, user, country, or distillery average rating were not as predictive as whiskey average rating. We then considered various other features that might be predictive, such as whiskey category, price, the whiskey's stated age, alcohol strength, and the vintage date. For category, we used a one-hot encoding scheme, since categories are classes and should not be assigned a numerical value. Similarly, for vintage, we also used a one-hot encoding scheme, because a specific vintage year might signify good whiskeys, but neighboring years might not necessarily do so. Since including both one-hot encodings will significantly increase the feature size, we optimized the model further by varying the regularization parameters to find the lowest validation RMSE.

Jaccard Similarity

Our second approach is to measure user-to-user similarity using Jaccard similarity. Jaccard similarity measures the similarity between finite sample sets by taking intersection over union of the sample sets. So, given a pair of a user and a whiskey (u, w), we would consider all training items w' that the user u has reviewed. Then, for each pair, we computed the Jaccard similarity between w and w', i.e., users (in the training set) who have reviewed w and users who have reviewed w'. Using a similarity score furnishes the model with some form of user-customization, especially when combined with other models.

Bag of Words

Our third approach is to make use of the review text in the reviews dataset to examine the correlation between text data and rating. Since not all users who give ratings to whiskeys also write reviews, we would only consider users in the review dataset (as opposed to the ratings dataset). First, we took the total review length by summing up the length of message, nose, taste, and finish fields for each review into a linear regression model. However, solely using this feature turns out to be not a good predictive variable as it yields RMSE of 5.654, roughly 40% worse than our baseline model.

Therefore, we tried to incorporate some form of sentiment analysis; we included the review texts themselves using the bag-of-words model. To build a dictionary of most frequently occuring words that factors out meaningless words, we removed capitalization, punctuation, and stop words. We then built the feature vector by counting the number of occurrences of the most common words across in the training set within a single review. After that, we further optimized the model by performing a grid search for the regularization parameter.

Hybrid Model

The three models we explored above perform different things and each of them has different strengths. Naturally, we attempted to create an even better model by combining the 3 alternative models above, so that their differing strengths can complement each other.

The feature combination model is capable of learning features quickly and obtained the lowest MSE out of all three alternatives; hence, this is the model we start building from. However, feature combination alone will predict the same rating for a given whiskey, even for very different users. The Jaccard similarity model aims to solve this by introducing a level of user-customization to the model. After that, we can further augment our prediction capability by analyzing the available textual data and incorporating the text semantics into the model through sentiment analysis. Once we find the most optimal hybrid model, we can further optimize that model by selecting regularization parameters. The results are reported in the Results section below.

Issues

We ran into scalability issues when computing Jaccard similarity scores for all ratings. The complexity of this operation is $O(r^*(u+w))$, where r is the number of ratings, u is the number of users, and w is the number of whiskeys. To overcome this issue, we tried to minimize (u+w) so as to not reduce r. We filtered out users who have rated only 1 whiskey as well as whiskeys that only had 1 rating from the datasets.

We also ran into overfitting problems when developing the bag-of-words model; the learned coefficients and MSE would occasionally rise to a very high figure. We overcame this problem by using ridge regression instead of linear regression, because the regularization will shrink spurious predictors that will potentially overfit the data. In addition, we also tried to mitigate overfitting by carefully choosing a dictionary size that is not too big, yet big enough to be predictive through a grid search. We noticed a decreasing pattern of RMSE starting from size of 1,000 until 3,000 as the lowest, then the RMSE increased onwards.

Literature

Our Dataset

We retrieved our datasets by scraping data from *www.whiskeybase.com*, the biggest resource of whiskey information in the world managed by whiskey enthusiasts since 2007. As we initially could not find reasonably large whiskey datasets with the relevant information needed for our predictive task, we decided to collect our own datasets from this whiskey database website. Since they do not provide any downloadable dataset, we created our own multithreaded python script utilizing BeautifulSoup and mechanize, http cookie jar, and requests libraries to extract the data we need.

Similar Studies

We found some similar whiskey dataset being used to study a different predictive task, as seen in the paper "Building a Multi-Agent Whisky Recommender System" by Torje Mjønes Coldevin[2]. He based his recommender system on several whiskey datasets taken from books and the internet, e.g. "Whisky Classified" book, "Maltwhisky: handbok" book, and *http://www.whisky-distilleries.info*. He aimed to build a classifier to identify whiskeys' taste classes by exploring different classification models such as Nearest Neighbor and Naive Bayes.

Coldevin's dataset contains many of the same features as our dataset. Several of these features are categorical features, such as whiskey category, cask type, etc. However, despite these similarities, we approached the categorical data in a different way; we encoded the categories using a one-hot encoding scheme to be able to pass it into a regression model, while Coldevin assigned a numerical label (an integer value) from 1 to 8, and used it as a feature to the non-regression classification models. Moreover, the general predictive task is different; Coldevin's predictive task is to classify whiskey tastes by using classification, whereas ours is to predict rating of whiskeys given by a user by using regression.

Furthermore, we also found a similar predictive task with different datasets being studied by Xu, Ke and Xixi Wang in their paper "Wine Rating Prediction"[3]. They scraped their data from WineEnthusiast with the goal of predicting rating points of wines based on the price, wine variety, and several winery location related information as the training features. As their study is similar to our work to predict a real-number value of rating, he also focused on exploring a variety of regression modelings such as basic linear regression, lasso regression, ridge regression, elastic net regression, and neural network. Their evaluation results concluded that ridge regression performs better than other models, and this aligns with our choice of model.

State-of-the-art Methods

According to the IEEE journal titled "Multi-Criteria Review-Based Recommender System-The State of the Art"[4], the state-of-the-art method that has been studied recently for multi-criteria review-based recommendation approaches uses user reviews to extract the criteria as a feature to the recommendation model. This approach can identify correlation between the users and items, and thus make the recommendation based on users' preferences instead of solely a single-criterion rating. The multi-criteria include total review polarity score, review terms, review feature/aspect/topic, total context, review comparative words, review emoticons, and review helpfulness. Applying these review elements to the recommended system also shows apparent improvement in performance as compared to other text mining techniques such as text and sentiment analysis.

This method is currently employed to study our type of data with ratings and reviews. We already implemented a part of the findings that highlighted the importance of incorporating multiple features instead of only relying on the overall rating. We also examined interactions between users and items using Jaccard similarity. We would further explore implicit values of the reviews as defined in the review and utilize them in future studies.

Results & Conclusion

Results

Combination of Features

By using the ridge regression model, we obtained the following RMSE on validation and testing sets for all 64 combinations of features. The following are some combinations of fields:

Feature	Val. RMSE	Testing RMSE
Offset	3.582	3.558
Category	3.557	3.532
Overall Rating of Whiskey	2.911	2.918
Category, Overall Rating of Whiskey	2.910	2.918
Category, Overall Rating of Whiskey, Strength	2.911	2.919
Category, Overall Rating of Whiskey, Price, Age	2.913	2.920
Category, Overall Rating of Whiskey, Vintage Age	2.917	2.928
Category, Overall Rating of Whiskey, Strength, Price, Age	2.914	2.922
Category, Overall Rating of Whiskey, Strength, Vintage Age	2.919	2.930
Category, Overall Rating of Whiskey, Strength, Price, Age, Vintage Age	2.918	2.929
Category, Overall Rating of Whiskey, Age, Vintage Age	2.917	2.927

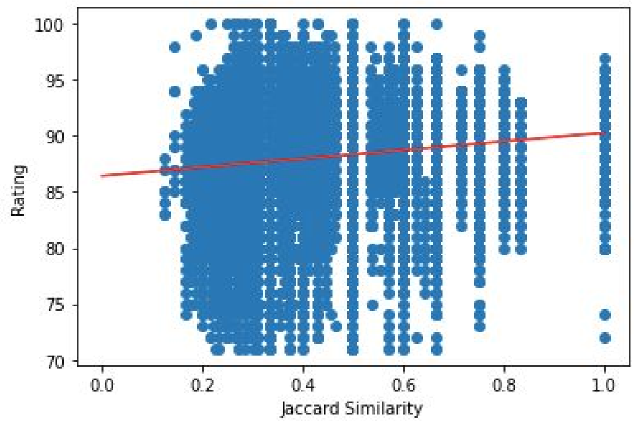
Based on all 64 RMSEs, we concluded that the meaningful features for this model are the overall rating of whiskeys and whiskey categories. This combination can reduce the RMSE to 2.910 on validation sets; including any other features increases the RMSE back up.

Overall rating of the whiskey is the single most important feature that makes up the bulk of the predictive power of the model. This is because users typically agree with other users regarding their rating of a whiskey; a good whiskey consistently gets an above average rating from all users. Category also adds predictive power because people might rate whiskeys of different categories differently. On the other hand, whiskeys' price, stated age, and strength did not add any predictive power (and in fact made the performance worse) because these features might be correlated with the overall rating. Having collinearity increases the variance of the error terms, making the model perform worse.

Jaccard Similarity

Feature	Val. RMSE	Testing RMSE
Jaccard Similarity (Mean)	3.582	3.594
Jaccard Similarity (Max)	4.134	4.263

If our model only operates purely based on user-to-user similarity, we beat the baseline by roughly 11% with RMSE of 3.582. However, Jaccard similarity on its own cannot beat average whiskey rating. This is because the similarity score only tells how similar a user is to another user, it cannot tell whether the whiskey that linked the two users is good or bad. The scatterplot above shows that, even with a perfect score of 1.0, the whiskey rating can still range from 75 to 95.



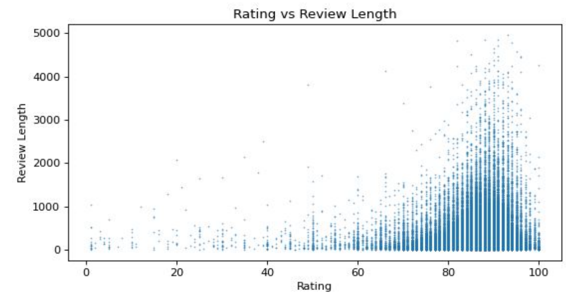
Furthermore, even if two users have exactly the same profile and taste preference, they might rate things differently. One user might give a 5 for good enough, whereas another might give 8 for good enough. This behavior is definitely not captured from just including the similarity score as the only feature.

Where this model excels is in providing personalization. Even though the predicted ratings might have a big error, this algorithm can identify other users that are similar to the specified user, and can recommend whiskeys based on which other whiskeys are rated the highest by the other users.

Bag of Words

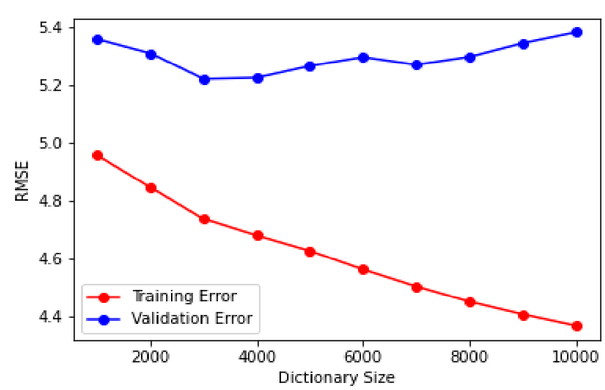
Feature	Val. RMSE	Testing RMSE
Review text length	5.863	5.654
Bag of Words + review text length	5.219	4.995

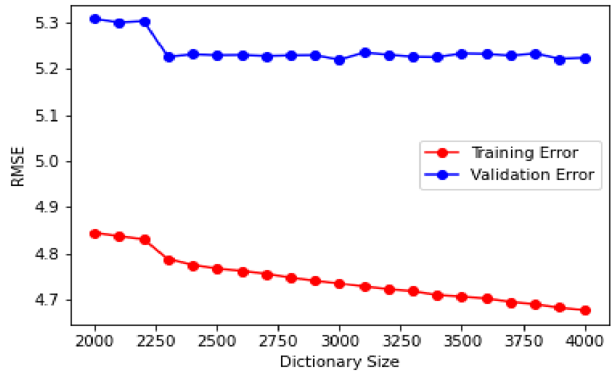
As expected, review length on its own is not enough, because in a big enough dataset, users are as likely to write short reviews for good whiskeys as they are to write long reviews. This can be seen in the scatterplot below.



Thus, we need to perform a deeper textual analysis to increase the predictive power. To analyze the meaning behind the words, we can perform a sentiment analysis using the bag-of-words technique. Using these as features, we gained an appreciable increase in performance, as evident in the 12% drop in RMSE.

We then compared different dictionary sizes ranging from 1,000 to 30,000 to find the size that yields the best performance. As a result, a dictionary size of 3,000 gives the best RMSE of 5.219 on validation sets and 4.995 on testing sets. The training errors and validation errors can be seen in the plot below as dictionary size increases.





Hybrid Model

Finally, we attempted to find the best combination of the 3 alternative models using ridge regression, and the summary of RMSE is as follows:

Feature	Val. RMSE	Testing RMSE
Category, Overall Rating of Whiskey	2.910	2.918
Overall Rating of Whiskey, Jaccard Similarity	2.912	2.919
Category, Overall Rating of Whiskey, Jaccard Similarity	2.909	2.917
Category, Overall Rating of Whiskey, Jaccard Similarity (C = 10)	2.906	2.911
Category, Overall Rating of Whiskey, Text Length, Bag of Words	4.259	4.428
Category, Overall Rating of Whiskey, Jaccard Similarity, Text Length, Bag of Words	4.258	4.427

As shown in the table, the combination of overall rating of whiskeys, category, and Jaccard similarity as variables in our ridge regression model produces the best RMSE of 2.909, significantly improves our baseline model by 28%. Then, to further improve its performance, we tried to find the best regularization constant from range 10^{-3} to 10^{3}. As a result, using the same combination with C = 10 produces slightly better RMSE of 2.906 on validation sets and 2.911 on testing sets.

Conclusions

We explored and optimized several regression models, namely feature combinations, Jaccard similarity, and bag-of-words. We then tried to combine all three models into one hybrid model with hopes that their strengths would complement each other. However, we soon found out that the bag-of-words model did not increase the predictive power of the model. Thus, we concluded that the best possible model considered so far is a model that incorporates Jaccard similarity and a combination of features of just the average whiskey rating and the whiskey's category. We incorporated these three features into a ridge regression model with a regularization constant of 10.

To further improve the quality of the results, we would like to explore other predictive regression modelings such as the non-linear kernel regression, lasso regression, and various types of neural networks in the future. We would also consider different sentiment analysis techniques on the review data by using the TF-IDF model instead of simple bag-of-words. Additionally, we would also investigate more on tuning more hyperparameters in our final model to have better results.

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